Note on Quantum Hall Effect

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April 09, 2026

Abstract

This is a note on QHE, based on David Tong's lecture notes.

Contents

1 Preparations

First we'll give some preparations for the discussion of the QHE. We'll start with some basic concepts in CMT and QM. I won't give a detailed explanation, yet just list out some important results as a reference for later use. Most contents are explained in [1].

1.1 Band Theory and Conductivity

1.1.1 Basic Concepts

In a crystalline solid, the periodic ionic potential causes the allowed electron energies to split into **energy bands**, separated by **band gaps**. The key results are:

> **Theorem 1.1** **Bloch's Theorem**
>
> The eigenstates of an electron in a periodic potential $V(\boldsymbol{r}+\boldsymbol{R}) = V(\boldsymbol{r})$ take the form
>
> $$\psi_{\boldsymbol{k}}(\boldsymbol{r}) = e^{i\boldsymbol{k}\cdot\boldsymbol{r}} u_{\boldsymbol{k}}(\boldsymbol{r}), \tag{1}$$
>
> where $u_{\boldsymbol{k}}$ shares the lattice periodicity $u_{\boldsymbol{k}}(r+R) = u_{\boldsymbol{k}}(r)$and $\boldsymbol{k}$ lives in the first Brillouin zone:
>
> $$\boldsymbol{k} \in \mathrm{BZ} = \left\{ \sum_{i=1}^{d} \alpha_i \boldsymbol{b_i} \mid 0 \leq \alpha_i < 1 \right\} \tag{2}$$
>
> where $\boldsymbol{b_i}$ are the reciprocal lattice vectors.

Solving the Schrödinger equation with Bloch boundary conditions yields a discrete family of dispersion relations $E_n(\boldsymbol{k})$, labelled by the band index n.

Conceptually, what we do is that we find a symmetry operator T which is the lattice translation operator, and we notice that it commutes with the Hamiltonian, thus we can find a common eigenstate of both H and T to classify the eigenstates into bands.

If the system has a finite number of length for example its a rectangular box L_1, L_2. And we impose suitable boundary conditions, then the allowed $\boldsymbol{k}$ values are discrete, we can calculate how many $\boldsymbol{k}$ states are there in each band, and we find that the number of states in each band is equal to the number of unit cells in the system:

$$N_k = N_c = \frac{L_1 L_2}{a_1 a_2} \tag{3}$$

where a_1, a_2 are the lattice constants. This means that each band can accommodate N_c electrons (ignoring spin).

Here is a rough sketch of the band structure:

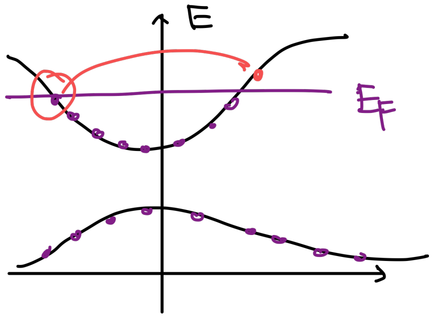

Figure 1: Band structure

Fermi energy and band filling. At $T = 0$, electrons fill all states up to the Fermi energy E_F.

1.1.2 Group Velocity

The group velocity of an electron in a band is given by the gradient of the energy dispersion:

$$v_n(\boldsymbol{k}) = \frac{1}{\hbar} \nabla_{\boldsymbol{k}} E_n(\boldsymbol{k}) \tag{4}$$

This means that the electron's velocity depends on its position in the Brillouin zone and the curvature of the energy band.

This quantity has a physical meaning: if we prepare a wave packet that is a superposition of Bloch states around a certain $\boldsymbol{k}$ point, then the wave packet will move with the group velocity $v_n(\boldsymbol{k})$.

1.1.3 Conductivity

Now let's discuss the conductivity of a material. The key results are:

- **Insulator / Semiconductor:** E_F lies inside a band gap; the valence band is completely full.
- **Metal:** E_F lies inside a band; there exists a Fermi surface.
- **Semimetal:** valence and conduction bands overlap slightly.

1.2 Landau Levels

1.2.1 Landau Gauge

Consider a electron moving in a 2D plane under a perpendicular magnetic field B. The Hamiltonian in Landau Gauge is:

$$H = \frac{1}{2m}\left(p_x^2 + \left(p_y + eBx\right)^2\right) \tag{5}$$

Then the energy spectrum is quantized into **Landau levels**:

- **Wave Functions**

$$\psi_{n,k}(x, y) \sim e^{iky} H_n(x + kl_B^2) e^{-(x+kl_B^2)^2/2l_B^2} \tag{6}$$

with $H_n \exp\left(-\frac{x^2}{2}\right)$ the n-th Harmonic oscillator wave function and $l_B = \sqrt{\hbar/eB}$ the magnetic length. And $n \in \mathbb{N}$ is the Landau level index, $k \in \mathbb{Z}$ is the momentum along y direction.

- **Energy Spectrum** The energy of the n-th Landau level is:

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega_B \quad \omega_B = e\frac{B}{m} \tag{7}$$

And each level have a degeneracy of

$$N_B = eBA/2\pi\hbar \quad n = \frac{B}{\Phi_0} \tag{8}$$

which is the number of flux quanta through the system. These energy levels can be drawn in the following CMT style:

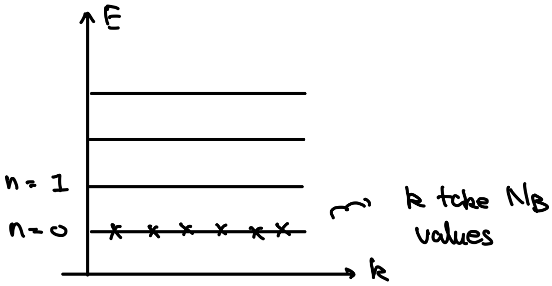

Figure 2: Landau levels

1.2.2 Landau Levels with Electric Field

If we add a uniform electric field E along x direction, then the Hamiltonian becomes:

$$H = \frac{1}{2m}\left(p_x^2 + \left(p_y + eBx\right)^2\right) - eEx \tag{9}$$

Then we solve the Schrödinger equation and find the energy spectrum is still quantized into Landau levels, but with a shift:

- **Wave Functions**

$$\psi(x, y) = \psi_{n,k}(x - mE/eB^2, y) \tag{10}$$

- **Energy Spectrum**

$$E_{n,k} = \hbar\omega_B\left(n + \frac{1}{2}\right) + eE\left(kl_B^2 - \frac{eE}{m\omega_B^2}\right) + \frac{m}{2}\frac{E^2}{B^2} \tag{11}$$

We now see that different k states are not degenerate anymore, and the landau levels are tilted in the energy-momentum space:

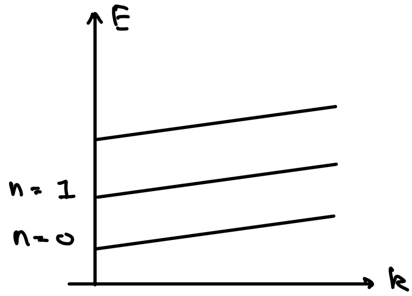

Figure 3: Landau levels with electric field

- **Group Velocity** If the wavefunction is a wave packet that is superposition of the landau level states with different k, then the group velocity of the wave packet is:

$$v_y = \frac{1}{\hbar}\frac{\partial E_{n,k}}{\partial k} = \frac{E}{B} \tag{12}$$

it means that the wave packet will drift along y direction with a velocity.

1.2.3 Symmetry Gauge

YL: [To be completed, and important for IQHE]

1.3 Berry Phase

When a quantum system has a parameter dependent Hamiltonian $H(\lambda_i)$ and the parameters vary adiabatically along a closed loop C in the parameter space, the system's ground state wave function acquires a geometric phase known as the **Berry phase**.

1.3.1 Abelian Berry Phase

Consider the ground state energy is always normalized to 0, no matter which parameter we choose, then after a close loop of parameter variation, the wave function will acquire a phase factor:

$$|\psi(\lambda_i)\rangle \to e^{i\gamma}\ |\psi(\lambda_i)\rangle \tag{13}$$

where γ is the Berry phase, which can be expressed as an integral of the Berry connection A_i along the loop

- **Berry Connection and Curvature** The Berry connection is defined as:

$$A_i(\lambda) = -i\langle n|\frac{\partial}{\partial \lambda^i}|n\rangle \tag{14}$$

where $|n(\lambda_i)\rangle$ is a series of designated eigenstates of the Hamiltonian with fixed phase choices. And the Berry curvature is defined as:

$$F_{ij}(\lambda) = \partial_i A_j - \partial_j A_i \tag{15}$$

notice that i, j are indices of the parameters, not the spatial coordinates.

- **Gauge Transformation** if we change the choise of phase of the designated eigenstates by a gauge transformation $|n(\lambda_i)\rangle \to e^{i\alpha(\lambda)}\ |n(\lambda_i)\rangle$, then the Berry connection transforms as:

$$A_i(\lambda) \to A_i(\lambda) + \partial_i \alpha(\lambda) \tag{16}$$

which is similar to the gauge transformation of the electromagnetic vector potential. However, the Berry curvature is gauge invariant.

- **Berry Phase** can be expressed as the integral of the Berry connection along the loop:

$$i\gamma = -i\oint_C A_i(\lambda) d\lambda^i \tag{17}$$

or equivalently as the integral of the Berry curvature over the area enclosed by the loop:

$$i\gamma = -i\oint_S F_{ij}(\lambda) dS^{ij} \tag{18}$$

- **Chern Number** In a parameter space if we want the berry phase to be well-defined, then the integral of the Berry curvature over a closed surface must be quantized in units of 2π:

$$C = \frac{1}{2\pi}\oint_S F_{ij}(\lambda) dS^{ij} \in \mathbb{Z} \tag{19}$$

this integer C is called the Chern number, which is a topological invariant of the system.

Remark

The quantization of the Chern number is a result of the fact that we want a well-defined holonomy of the Berry connection. This derivation in fact can be generalized to any number of parameters, integrating such a curvature over a closed manifold will give a topological invariant, which is the Chern number.

YL: [If time allows, I can go and see it in a general construction]

1.3.2 Non-Abelian Berry Phase

When the system has degenerate ground states, the Berry phase becomes non-Abelian. We can see this from the fact that the designated eigenstates should span the degenerate subspace:

$$|n(\lambda_i)\rangle \to \{|n_a(\lambda_i)\rangle\}_a \tag{20}$$

Then we generalize the definition of Berry connection to a matrix-valued one.

- **Non-Abelian Berry Connection and Curvature** The non-Abelian Berry connection is defined as:

$$A_i(\lambda)_{ab} = -i\langle n_a|\frac{\partial}{\partial\lambda^i}|n_b\rangle \tag{21}$$

and the non-Abelian Berry curvature is defined as:

$$F_{ij}(\lambda) = \partial_i A_j - \partial_j A_i - i[A_i, A_j] \tag{22}$$

- **Gauge Transformation** if we change the choice of basis of the degenerate subspace by a gauge transformation $|n_a(\lambda_i)\rangle \to U_{ab}(\lambda)\ |n_b(\lambda_i)\rangle$, where $U(\lambda)$ is a unitary matrix, then the non-Abelian Berry connection transforms as:

$$A_i(\lambda) \to U(\lambda)A_i(\lambda)U^\dagger(\lambda) + i(\partial_i U(\lambda))U^\dagger(\lambda) \tag{23}$$

Then we notice that the berry connection is a $U(N)$ gauge field, that take valve of $u(N)$ Lie algebra. And the non-Abelian Berry curvature transforms as:

$$F_{ij}(\lambda) \to U(\lambda)F_{ij}(\lambda)U^\dagger(\lambda) \tag{24}$$

- **Berry Holonomy** The non-Abelian Berry phase (in fact its a matrix) can be expressed as a path-ordered exponential of the Berry connection along the loop:

$$U = P\exp\left(-i\oint_C A_i(\lambda)d\lambda^i\right) \tag{25}$$

1.3.3 Spectrum Flow

2 Integer Quantum Hall Effect

Now we're prepared to discuss the integer quantum Hall effect (IQHE). The IQHE is a quantum phenomenon observed in two-dimensional electron systems subjected to low temperatures and strong magnetic fields.

2.1 Standard QHE

The standard IQHE occurs in a two-dimensional electron gas (2DEG) confined to a plane and subjected to a perpendicular magnetic field. This section we will discuss the explanation of IQHE in this framework.

2.1.1 Experimental Observations

2.1.2 Quantization of Hall Conductance

2.1.3 Robustness of Hall Conductance

2.2 TKNN Invariant

Apart from the standard set up of IQHE. There is a more general framework of models that can exhibit a quantized Hall conductance σ_{xy} even ones without a external magnetic field.[1] The underlying reason for this quantization can be illustrated in a same manner, which is the TKNN invariant.

2.2.1 Kubo Formula

2.2.2 TKNN Formula

2.3 Example: Chern Insulator Models

This is a series of model that can have nontrivial TKNN invariant and thus exhibit quantized Hall conductance without external magnetic field.

The most famous one is the Haldane model, which is a tight-binding model on a honeycomb lattice with complex next-nearest neighbor hopping. Here we might just discussed another simpler model, the Qi-Wu-Zhang (QWZ) model [3].

2.4 Example:

[1]This phenomenon of QHE without Magnetic field is called the quantum anomalous Hall effect (QAHE). Btw, this first observation is made in 2013 by a THU team [2]

Bibliography

[1] D. Tong, "Lectures on the Quantum Hall Effect." Accessed: Mar. 31, 2026. [Online]. Available: http://arxiv.org/abs/1606.06687

[2] C.-Z. Chang *et al.*, "Experimental Observation of the Quantum Anomalous Hall Effect in a Magnetic Topological Insulator," *Science*, vol. 340, no. 6129, pp. 167–170, Apr. 2013, doi: 10.1126/science.1234414.

[3] X.-L. Qi, Y.-S. Wu, and S.-C. Zhang, "Topological quantization of the spin Hall effect in two-dimensional paramagnetic semiconductors," *Physical Review B*, vol. 74, no. 8, Aug. 2006, doi: 10.1103/physrevb.74.085308.